

ECOLOGY

CO, NH₃, NO₂, PM_{2.5}, pressure, temperature
└──────────┴──────────┘
↓
Account for AQI

* Traditional Air quality monitoring methods

- └─ high capital investment
- └─ high maintenance
- └─ Limited spatial resolution

* MICS - 6814

- For CO, NO₂, NH₃

Why:- Compact & low power, cost effective, fast response

Disadvantages: ① Outputs are raw values

↓
Calibration is required

② Useful for trend monitoring

It is of Metal oxide Semiconductor (MOS) type

* PM_{2.5} GP2Y1010AUF⁰

- PM_{2.5}

Why:- Low cost, Compact & less weight

- Analog output

Disadvantage - Calibration is needed

* BMP 280 Sensors :

- Pressure temperature & altitude

Why:- high accuracy, low power, less size

Disadvantages:-

① No humidity

② Altitude estimation varies with weather

* ESP-32 - Microcontroller with built in wifi & bluetooth capabilities

* NEO UBLOX GM GPS

- GPS co-ordinates measurement

Why:- Good Accuracy, low power consumption, connection compatibility

* LoRa Module

- Data transfer

Why:- Long range communication, low power consumption

Disadvantages:-

- Low data transfer rate

- Limited packets of size

- Antenna quality management

- ① MICS - 3 analog output $\rightarrow$ ESP32
- ② BMP280 - I2C protocol through SDA & SCL pins
Inter-Integrated Serial serial
Circuit Data clock
- ③ PM 2.5 GP2Y1010A0F - Analog output - ESP32
- ④ LoRa module - digit I/O pins SPI
Serial peripheral Interface
- ⑤ NEO UBLOX GPS - UART using Tx & Rx pins
 $\downarrow$ $\downarrow$
Transmit Receive
- ⑥ Reference grade Air Quality monitoring instrument
- high precision device used by govt agencies,
research institutions.

AQI Calculation:

$(CP(B))$
(6500)

Calculation:
24hrs average value - NO_2 , SO_2 , $\text{PM}_{2.5}$, PM_{10} etc.

8 hrs average value - CO_2 & O_3

చ) 8 values తీసుకోవాల్సిన అవసరం లేదు-

① Minimum of 3 correct answers required

PM-10 / PM2.5 மைய கட்டி

Sub-index calculate ≥ 12 minimum of 16
hours good

AQI	Category
0-50	Good
51-100	Satisfactory
101-200	Moderate
201-300	Poor
301-400	Very poor
401-500	Severe

* Standards of $PM_{2.5}$, NO_2 and CO are listed in report

$CO - \mu g/m^3$ - 8hrs average

$NO_2 - \mu g/m^3$ - 24hrs average

$PM_{2.5} - \mu g/m^3$ - 24hrs average

As per NAAQS (National Ambient Air Quality Standards)
- 2009

Permissible limits are

(i) $PM_{2.5}$ - 24-hours average - $60 \mu g/m^3$

Annual average - $40 \mu g/m^3$

(ii) NO_2 - 24-hours average - $80 \mu g/m^3$

Annual average - $40 \mu g/m^3$

(iii) CO - 24-hours average - $2 mg/m^3$

1-hours average - $4 mg/m^3$